

1 Task Package

Example Task (Multimodal)

Task Instruction: Cook Beef Wellington

1. Find official YouTube tutorial
2. Watch video
3. Capture ≥3 key-step screenshots
4. Read my_fridge.jpg
5. Generate:
 - shopping_list.md
 - cooking_steps.md

Provided Resources

- 🌐 Live web access (e.g., YouTube)
- 📁 Empty workspace
- 🖼️ Initial files (e.g., my_fridge.jpg)
- 💻 Installed tools & services (Browser, GUI Apps, Skills etc.)

Example Fridge Photo



Hidden Resources

- 🔒 References and ground truth
- 🔒 Evaluation rule

2 Executor Agent (in Docker)



Agent Framework (e.g., OpenClaw/Nanobot/EDICT)

Example Trajectory (visible)

Step 1	Search YouTube "Gordin Ramsay Beef Wellington"	🌐
Step 2	Play video and watch cooking process	📺
Step 3	Capture screenshot: searing beef	🖼️
Step 4	Capture screenshot: wrapping pastry	🖼️
Step 5	Capture screenshot: egg wash	🖼️
Step 6	Read my_fridge.jpg	🖼️
Step 7	Compare ingredients with fridge photo	🖼️
Step 8	Write and save shopping_list.md	📄
Step 9	Write and save cooking_steps.md	📄

Produced Artifacts (visible)



Screenshot_step1.png



Screenshot_step2.png



Screenshot_step3.png



Shipping_list.md



Cooking_steps.md

Natural Language Feedback
from User Simulator

3A Supervisor Agent

Inputs

- Executor trajectory & artifact
- Hidden References & ground truth
- Evaluation rule

Hidden Evaluation Rubric

ID	Checkpoint (weight)	Status
C1	Official video source (0.10)	✅ pass
C2	≥3 valid cooking screenshots (0.20)	✅ pass
C3	Describe ≥3 steps & links files (0.15)	✅ pass
C4	Fridge photo grounding (0.15)	❌ fail
C5	Missing-ingredient accuracy (0.30)	❌ fail
C6	Both MD files exist & complete (0.10)	❌ fail

Coarse progress signal

Score: 0.65

Status:

complete but failed

Recoverable: **True**

Verdict: **Continue**



INFORMATION FIREWALL (No Leakage)

3B User Simulator

Inputs

- Executor trajectory & artifact (same as supervisor)
- Coarse progress signal

User Simulator (LLM-based):

- Simulate a real user who checks the results and provides natural feedback



Verify the screenshots against the video, align filenames and steps, and ensure the shopping list includes all items not visible in the fridge.

4 Termination Conditions (per task)

- ✅ **Pass**: Final score ≥ threshold (e.g., 0.9).
- 🕒 **Continue**: task not passed but recoverable AND follow-ups remain (e.g., max 3 total).
- ❌ **Fail**: task not passed and unrecoverable OR task not passed and no follow-ups remain.
- ⌚ **Timeout**: executor fails to complete within the time limit (e.g., 30 minutes).

5 Final Metrics (by supervisor)

Supervisor makes the final judgement using evaluation rule

Metric 1: PR (Pass Rate)

Task-level: pass / fail

Metric 2: AS (Average Score)

Step-level: aggregated score

